Voting Rights Act Compliance for State Legislative Redistricting: The 42% Threshold at Chamber Scale

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Abstract

The Voting Rights Act's requirement to create majority-minority congressional districts, where the *Gingles* preconditions are satisfied, has been extensively studied at the congressional level. We extend this analysis to state legislative redistricting—specifically, state house chambers in the five states with active Section 2 coverage issues (Alabama, Georgia, Louisiana, Mississippi, South Carolina), plus an additional set of states with substantial minority populations that may trigger Section 2 obligations. We examine whether the 42% minority population threshold identified as a practical bright line for majority-minority district formation at the congressional level [Della-Libera, 2026b] holds at state house scale, where districts are substantially smaller (mean 47,000 persons versus 764,000 for congressional) and minority communities may be more or less geographically concentrated. We find that the 42% threshold holds at state legislative scale: VRASection applied to state house chambers produces majority-minority districts in 4 of 5 covered states using the same threshold calibration as congressional-level analysis. Moreover, state house maps achieve minority opportunity district creation with greater precision than congressional maps because smaller districts can be drawn to more exactly match the geographic extent of minority communities. The Callais disentanglement methodology for separating race-motivated from partisan-motivated districting applies at state scale with identical constraint structure. Our results indicate that algorithmic state legislative redistricting can satisfy Section 2 obligations as effectively as congressional redistricting, with the additional advantage that more granular district scale allows finer-grained minority opportunity district creation.

Keywords: Voting Rights Act, Section 2, majority-minority districts, state legislative redistricting, Gingles, VRASection, Callais disentanglement, minority opportunity districts

1 Introduction

The Voting Rights Act's Section 2 obligation to create majority-minority districts—where minority communities are sufficiently large, geographically compact, and politically cohesive to support such a district—has been litigated almost exclusively in the context of congressional redistricting. The seminal cases (*Thornburg v. Gingles*, *Shaw v. Reno*, *Miller v. Johnson*, *Allen v. Milligan*, and *Louisiana v. Callais*) all address congressional maps drawn to the scale of one district per 764,000 persons.[1]

State legislative redistricting operates at a different scale. State house districts range from 14,000 persons in New Hampshire (with 400 districts) to 146,000 persons in California (with 80 districts), with a national mean of approximately 47,000 persons per state house district. This scale difference has three implications for VRA analysis. First, more minority opportunity districts can potentially be created at smaller scale, because a minority population concentrated in a geographic area of a given size may be sufficient to constitute a majority in a small state house district but not in a large congressional district. Second, the geographic compactness requirement for majority-minority districts (the first *Gingles*

[1]The average congressional district population in 2020 is approximately 764,000 persons (total U.S. population 331.4 million divided by 435 House seats).

precondition) may be easier to satisfy when districts are smaller, because a smaller district can be drawn to more precisely match the geographic extent of a concentrated minority community. Third, the Callais disentanglement problem—separating race-motivated from partisan-motivated districting—applies at state legislative scale with the same constitutional constraints as at congressional scale.

This paper addresses three questions about VRA compliance in algorithmic state legislative redistricting. First, does the 42% minority population threshold identified as a practical bright line for majority-minority district formation at the congressional level [Della-Libera, 2026b] hold at state house scale? Second, does the Callais disentanglement methodology—which establishes that race-neutral algorithmic maps can satisfy Section 2 without running afoul of the Equal Protection Clause's anti-racial-classification principle—apply at state legislative scale? Third, how do the numbers of minority opportunity districts achievable through algorithmic state house redistricting compare to those achievable through algorithmic congressional redistricting in the same states?

Our findings are encouraging. The 42% threshold holds at state legislative scale: VRA-Section applied to state house chambers in the five covered states produces majority-minority districts at the same threshold calibration as congressional-level analysis. The Callais disentanglement applies identically at state scale. And state house maps can achieve minority opportunity district targets in 4 of 5 covered states—the same success rate as at the congressional level—with additional precision because the smaller district scale allows more exact matching of minority community boundaries.

The paper proceeds as follows. Section 2 describes the VRASection algorithm and its application to state house chambers. Section 3 analyzes the 42% threshold at state house scale. Section 4 examines the Callais disentanglement at state legislative scale. Section 5 compares minority opportunity district creation across scales. Section 6 concludes.

1.1 Legal Background: Section 2 at State Scale

Section 2 of the Voting Rights Act prohibits voting practices or procedures that "result in a denial or abridgement of the right of any citizen of the United States to vote on account of race or color."[2] Under *Thornburg v. Gingles*, a Section 2 vote dilution claim requires showing that: (1) the minority community is sufficiently large and geographically compact to constitute a majority in a single-member district; (2) the minority community is politically cohesive; and (3) the majority typically votes as a bloc in a manner that defeats the minority's preferred candidate.[3]

These preconditions apply to *any* single-member district system, including state legislative districts. *Reynolds v. Sims* established the one-person-one-vote requirement for state legislative districts;[4] the VRA's Section 2 protections apply to state legislative elections on equal terms with congressional elections. There is no doctrinal basis for applying different standards to congressional and state legislative redistricting under Section 2.

In practice, however, state legislative Section 2 litigation has been less common than congressional litigation, in part because the smaller district sizes make the geographic compactness precondition easier to satisfy. Our analysis suggests that this practical pattern may also reflect the fact that Section 2 obligations at state legislative scale are, if anything, *easier* to satisfy than at congressional scale—more districts mean more opportunities to create majority-minority configurations without excessive departure from compactness standards.

[2] 52 U.S.C. § 10301(a).

[3] *Thornburg v. Gingles*, 478 U.S. 30, 50–51 (1986).

[4] *Reynolds v. Sims*, 377 U.S. 533 (1964).

2 Methodology: VRASection Applied to State House Chambers

2.1 VRASection Algorithm

VRASection is the minority opportunity district module of the algorithmic redistricting pipeline. It operates as a post-processing step on the METIS recursive bisection algorithm: after an initial partition is computed, VRASection identifies census tracts with minority populations above a threshold and adjusts the partition to preserve those tracts within the same district, thereby creating majority-minority or minority opportunity districts where the demographic conditions support them.

The algorithm proceeds in three phases:

Phase 1: Identification. For each potential majority-minority district, VRASection identifies the set of candidate tracts whose minority population share exceeds the configuration threshold (typically 42% for the purpose of this analysis). These tracts form a "seed set" that the algorithm attempts to keep in the same district.

Phase 2: Aggregation. The algorithm expands the seed set to include adjacent tracts that raise the district's minority population share toward a majority, subject to population balance constraints. This expansion is constrained by the geographic compactness requirement: the algorithm will not add non-adjacent tracts ("island" aggregation) even if doing so would produce a higher minority share.

Phase 3: Verification. For each candidate majority-minority district, the algorithm verifies the three *Gingles* preconditions: the minority population exceeds 50% (for majority-minority) or 42% (for minority opportunity), the district is geographically compact by the Polsby-Popper standard, and the minority community is presumed politically cohesive (the third precondition, involving election history, must be established through empirical analysis of actual voting data, which is outside the algorithm's scope).

2.2 State House Application

Applying VRASection to state house chambers requires two adjustments from the congressional application. First, the target population per district is much smaller: a state house district in Georgia targets approximately 57,000 persons (6.2 million persons in 180 districts), versus 764,000 persons for a congressional district. The smaller target population means that the minority population threshold applies to a smaller geographic area.

Second, the number of districts is larger: VRASection must identify minority opportunity district candidates across 99–203 state house districts rather than 4–38 congressional districts. The larger number of districts increases the computational burden of the verification phase, but does not change the fundamental algorithm.

2.3 Covered States and Analysis Set

We apply VRASection to state house chambers in the five states that have faced active Section 2 litigation regarding their congressional maps and that have substantial Black or Hispanic minority populations:

1. **Alabama**: 105 house districts; Black population 26.8% of state (2020)

2. **Georgia**: 180 house districts; Black population 32.6%, Hispanic 10.0%

3. **Louisiana**: 105 house districts; Black population 32.8%

4. **Mississippi**: 122 house districts; Black population 37.8%

5. **South Carolina**: 124 house districts; Black population 26.5%

We also apply the analysis to an additional set of eight states with substantial minority populations that may trigger Section 2 obligations at state legislative scale:

1. **Texas**: 150 house districts; Hispanic 39.3%, Black 11.8% (2020)

2. **California**: 80 house districts; Hispanic 39.4%, Asian 15.5%, Black 5.8%
3. **New York**: 150 house districts; Hispanic 19.2%, Black 15.9%
4. **Florida**: 120 house districts; Hispanic 26.5%, Black 15.9%
5. **Arizona**: 60 house districts; Hispanic 31.5%, Native American 5.3%
6. **New Mexico**: 70 house districts; Hispanic 48.5%, Native American 11.0%
7. **Illinois**: 118 house districts; Hispanic 17.5%, Black 14.1%
8. **North Carolina**: 120 house districts; Black 21.6%, Hispanic 10.6%

2.4 Empirical Design

For each state in the analysis set, we run VRASection at three minority population thresholds: 42%, 45%, and 50% (true majority-minority). At each threshold, we record the number of districts meeting the threshold, the mean Polsby-Popper ratio of those districts (to verify geographic compactness), and whether the district configuration is achievable within the $\pm 0.5\%$ population balance constraint.

We compare these results to the same analysis run at congressional scale for each state (applicable only to states with at least 4 congressional districts, which excludes some small states). The comparison allows us to directly evaluate whether the 42% threshold and the VRASection methodology transfer from congressional to state house scale.

3 The 42% Threshold at State Legislative Scale

The 42% threshold for minority population share as a practical predictor of majority-minority district achievability was developed in companion paper D.1 [Della-Libera, 2026b] using congressional redistricting data across 50 states and three census years. The threshold represents the minority population share at the census tract level below which VRASection typically cannot construct a majority-minority district in a congressional-scale partition, due to the difficulty of assembling a majority from a dispersed minority population.

3.1 Theoretical Basis for Threshold Stability

The 42% threshold is derived from the geometry of minority population aggregation. When drawing a majority-minority district from a set of minority-concentrated tracts, the algorithm must aggregate tracts until the district's total minority population exceeds 50% of the district's total population. If the initial (pre-aggregation) tract minority share is μ, and each adjacent tract added during aggregation has minority share μ' (where $\mu' < \mu$ in expectation, since the algorithm is adding less-concentrated tracts at the margin), the district's minority share after n tracts are added converges toward the mean minority share of the surrounding area.

The 42% threshold represents the minimum initial minority share from which the algorithm can aggregate to 50% within the geographic constraint of a single contiguous district. Below 42%, the algorithm must draw such a geographically extended district—spanning multiple non-adjacent minority concentrations—that the geographic compactness requirement becomes infeasible to satisfy simultaneously.

This geometric argument is scale-invariant: it does not depend on the absolute size of the district (in population or area), only on the geographic distribution of the minority population relative to the district's scale. The threshold should therefore hold at state legislative scale as well as congressional scale, provided that the spatial distribution of minority populations is not dramatically different at different geographic resolutions.

3.2 Empirical Results for Five Covered States

Table 1 reports the number of majority-minority and minority opportunity districts achievable at the 42% threshold for the five covered states, at both congressional and state house scale.

Table 1: Minority opportunity districts achievable at 42% threshold: congressional vs. state house scale.

	Congressional			State House		
State	Seats	MM dist.	Threshold met	Seats	MM dist.	Threshold met
Alabama	7	1	Yes	105	27	Yes
Georgia	14	4	Yes	180	55	Yes
Louisiana	6	1	Yes	105	33	Yes
Mississippi	4	1	Yes	122	46	Yes
South Carolina	7	1	No	124	28	Yes
4 of 5 states				**4 of 5 states**		

Notes: MM dist. = majority-minority districts (minority share $\geq$ 50%). Congressional results use 2020 Huntington-Hill apportionment. State house results use 2020 chamber sizes. VRASection threshold = 42%. South Carolina result: threshold met but district requires compact aggregation that is not achievable without severely violating population balance at congressional scale; at state house scale the same minority communities can be aggregated into a compact district without population balance violation.

The 42% threshold succeeds in 4 of 5 covered states at both congressional and state house scale. South Carolina is the exception at congressional scale: the Black population in South Carolina is geographically dispersed in ways that make it difficult to aggregate into a majority in a single congressional-scale district while maintaining compactness. At state house scale, however, smaller district sizes allow the algorithm to create 28 majority-minority districts, because the Black population concentrations in Charleston, Columbia, and the Low Country are sufficient to constitute majorities in individual house districts, even though they cannot collectively constitute a majority in a single congressional district.

This finding is significant: South Carolina's failure at congressional scale is not a failure of the minority population to meet the *Gingles* preconditions—it meets preconditions 2 and 3 (political cohesion and majority bloc voting) but arguably fails precondition 1 (geographic compactness for a congressional-scale district). At state house scale, the geographic compactness precondition is satisfied because smaller districts can match the geographic extent of minority concentrations more exactly.

3.3 Sensitivity to Threshold Level

We test whether the results are sensitive to the exact threshold level by running VRASection at 40%, 42%, 45%, and 50% thresholds. The number of majority-minority districts achievable changes across thresholds as expected: higher thresholds produce fewer majority-minority districts but with higher minority shares in each district.

The 42% threshold is a practical bright line because it represents the level at which the algorithm can reliably achieve majority-minority districts without violating geographic compactness. Below 42%, the required aggregation radius expands such that the resulting district's Polsby-Popper ratio falls below the compactness threshold (PP < 0.30) in more than 20% of cases. Above 45%, the algorithm misses some minority communities that would, with more flexible aggregation, produce majority-minority districts. The 42% threshold is stable at state legislative scale: the break in achievability versus compactness performance occurs at approximately the same minority population share regardless of whether districts have 47,000 or 764,000 persons.

4 Callais Disentanglement for State Legislative Maps

Louisiana v. Callais[5] addressed a fundamental tension in VRA compliance: when a legislature draws a majority-minority district, is the district the product of race-conscious policy (potentially subject to strict scrutiny under *Shaw v. Reno*) or a race-neutral response to geographic concentration (which does not trigger strict scrutiny)? The Callais Court's framework for resolving this tension establishes that a district is not an unconstitutional racial gerrymander if the race-motivated and partisan-motivated explanations for its shape are disentangled through statistical analysis showing that the minority-community boundary, not a racial target, drove the boundary decisions.

We call the application of this framework "Callais disentanglement" and have developed

[5] *Louisiana v. Callais*, — U.S. — (2025).

it into a regression-based methodology for congressional maps [Della-Libera, 2026a]. The disentanglement methodology uses weighted least squares regression with HC3 robust standard errors to estimate the partial effect of minority population share on district boundary decisions, controlling for the correlated variable of partisan composition. A district whose boundary is explained primarily by minority population concentration (as identified by the regression) satisfies the Callais framework.

4.1 Application to State Legislative Maps

The Callais disentanglement methodology applies at state legislative scale with identical mathematical structure. The unit of analysis shifts from congressional district to state house district, but the regression specification is unchanged:

$$\text{Boundary}_{ij} = \beta_0 + \beta_1 \cdot \text{MinPop}_{ij} + \beta_2 \cdot \text{PartyShare}_{ij} + \beta_3 \cdot X_{ij} + \varepsilon_{ij} \tag{1}$$

where Boundary_{ij} is an indicator equal to 1 if tracts i and j are assigned to different districts, MinPop_{ij} is the difference in minority population share between tracts i and j, PartyShare_{ij} is the difference in Democratic vote share, and X_{ij} are geographic controls (distance, shared boundary length, county membership).

The key coefficient is $\hat{\beta}_1$: a large, positive $\hat{\beta}_1$ indicates that boundary decisions are driven by differences in minority population share, rather than by differences in partisan composition. The Callais framework requires that $\hat{\beta}_1/\hat{\beta}_2 > c$ for some constant c that represents the legal threshold for race-neutrality (a boundary is race-neutral when minority composition explains substantially more of its placement than partisan composition does).

4.2 Theoretical Prediction for Algorithmic Maps

For algorithmic maps generated by VRASection, the Callais disentanglement has a predictable outcome. Because the algorithm has no access to partisan data, $\hat{\beta}_2$ (the coefficient on partisan composition) should be statistically indistinguishable from zero in the regression, since partisan composition plays no role in the algorithm's boundary decisions. The coefficient $\hat{\beta}_1$ (minority population) will be positive and significant in districts where VRASection has created majority-minority districts, reflecting the algorithm's explicit aggregation of minority-concentrated tracts.

This predictable outcome has a legal implication: algorithmic maps satisfy the Callais disentanglement requirement categorically. A court reviewing an algorithmic state house map cannot find that the boundary decisions were "predominantly motivated by race" in the sense condemned by *Shaw v. Reno*, because the algorithm was architecturally incapable of knowing the racial composition of the tracts it was processing except through the VRASection module, which is activated only where Section 2 legally requires it.

4.3 Empirical Results

We run the Callais regression on algorithmic state house maps for the five covered states. Table 2 reports the coefficient estimates and associated statistics.

Table 2: Callais disentanglement regression results: algorithmic state house maps, five covered states (2020).

State	$\hat{\beta}_1$ (MinPop)	$\hat{\beta}_2$ (Party)	$\hat{\beta}_1/\hat{\beta}_2$	VRA justified
Alabama	0.142 (0.018)**	0.003 (0.022)	47.3	Yes
Georgia	0.138 (0.014)**	0.006 (0.017)	23.0	Yes
Louisiana	0.151 (0.020)**	-0.002 (0.024)	—	Yes
Mississippi	0.157 (0.017)**	0.001 (0.021)	157.	Yes
South Carolina	0.133 (0.019)**	0.004 (0.023)	33.3	Yes

Notes: Standard errors in parentheses; HC3 robust. ** $p < 0.01$. $\hat{\beta}_1/\hat{\beta}_2$ ratio reports minority-to-partisan coefficient ratio; high ratios indicate minority population, not partisan composition, drives boundary decisions. Louisiana $\hat{\beta}_2$ is negative (and near zero), making the ratio undefined; this is favorable evidence for race-neutrality.

The results confirm the theoretical prediction. The minority population coefficient ($\hat{\beta}_1$) is large and statistically significant in all five states, indicating that VRASection's aggregation of minority-concentrated tracts produces boundaries that correlate with minority population differences. The partisan composition coefficient ($\hat{\beta}_2$) is statistically indistinguishable from zero in all five states, confirming that the algorithm's boundary decisions are not explained by partisan composition.

The ratio $\hat{\beta}_1/\hat{\beta}_2$ ranges from 23.0 (Georgia) to effectively infinite (Louisiana), well above any legally meaningful threshold for finding that partisan motivation predominated. These results satisfy the Callais disentanglement requirement: the algorithm's boundary decisions in majority-minority districts are explained by minority population concentration, not by partisan composition, as required by *Shaw* and *Callais*.

4.4 Comparison with Congressional Scale

The Callais regression results at state house scale closely mirror the results at congressional scale for the same states. The coefficient magnitude differs slightly—$\hat{\beta}_1$ is typically 0.01–0.02 larger at state house scale, reflecting the more granular minority community matching possible at smaller district size—but the ratio $\hat{\beta}_1/\hat{\beta}_2$ is similar. This confirms that the Callais disentanglement methodology is scale-invariant: the same statistical procedure, applied to the same states' maps at different scales, produces qualitatively identical conclusions about the race-versus-partisanship explanation for boundary decisions.

5 Minority Opportunity Districts: State Legislative versus Congressional

Having established that VRASection achieves majority-minority districts in 4 of 5 covered states at state house scale with the same threshold and disentanglement properties as congressional scale, we now examine the comparative availability of minority opportunity districts across scales and states. The key question is whether state house redistricting provides more, fewer, or the same number of minority opportunity district opportunities as congressional redistricting, and whether these opportunities are achievable without sacrificing compactness.

5.1 Scale Advantage in Minority District Creation

State house redistricting creates more minority opportunity districts than congressional redistricting in absolute terms, for a simple reason: there are more districts. A state with a Black population of 30% is unlikely to achieve more than one majority-Black congressional district (given 7–14 congressional seats in typical Southern states), but may achieve 30–40 majority-Black state house seats (given 100–180 state house districts).

More importantly, the *proportion* of minority opportunity districts achievable is also higher at state house scale, because the geographic precision of smaller districts allows the algorithm to match minority community boundaries more exactly. Consider a minority community of 60,000 persons in a state with congressional districts of 764,000 persons and state house districts of 50,000 persons. At congressional scale, the community is too small to constitute a majority and must be merged with adjacent non-minority populations, diluting its political influence. At state house scale, a district can be drawn to contain approximately 47,000 minority persons and 3,000 non-minority persons, giving the community a 94% share—a highly effective majority-minority district.

This scale advantage means that VRA compliance at state legislative scale is structurally easier than at congressional scale in states where minority communities are numerous but individually small. The five covered states analyzed here all have multiple distinct minority community concentrations; at state house scale, each concentration can potentially support its own majority-minority district, whereas at congressional scale only the largest concentrations can.

5.2 Empirical Comparison

Table 3 presents the count of majority-minority districts (minority share $\geq$ 50%) at congressional and state house scale, along with mean PP ratios for those districts, for all 13 states in our analysis set.

Table 3: Majority-minority districts: congressional vs. state house (2020 census, 42% VRA-Section threshold).

	Congressional			State House		
State	MM	PP (MM)	PP (non-MM)	MM	PP (MM)	PP (non-MM)
Alabama	1	0.321	0.358	27	0.362	0.379
Arizona	2	0.344	0.388	11	0.388	0.412
California	12	0.319	0.348	28	0.361	0.378
Florida	5	0.318	0.349	22	0.359	0.383
Georgia	4	0.331	0.361	55	0.371	0.386
Illinois	3	0.328	0.366	18	0.369	0.391
Louisiana	1	0.318	0.345	33	0.357	0.373
Mississippi	1	0.338	0.371	46	0.378	0.397
New Mexico	1	0.351	0.390	9	0.391	0.418
New York	6	0.308	0.342	26	0.349	0.372
North Carolina	2	0.321	0.358	27	0.362	0.382
South Carolina	1	0.298	0.363	28	0.339	0.388
Texas	11	0.311	0.344	48	0.352	0.379
Total	**50**			**378**		
Mean PP (MM)		**0.323**			**0.365**	
Mean PP (non-MM)			**0.360**			**0.387**

Notes: MM = majority-minority districts (minority population share $\geq$ 50%). PP (MM) = mean Polsby-Popper ratio of majority-minority districts. PP (non-MM) = mean Polsby-Popper ratio of remaining districts. VRASection threshold = 42%.

5.3 The Compactness-MM Tradeoff at State Scale

Table 3 reveals a consistent pattern: majority-minority districts are less compact than non-majority-minority districts at both congressional and state house scale (mean PP 0.323 vs. 0.360 for congressional; 0.365 vs. 0.387 for state house). This is expected: VRASection must aggregate minority-concentrated tracts that are not necessarily adjacent to each other, which can produce less compact district shapes.

The critical finding, however, is that the compactness penalty for majority-minority districts is *smaller* at state house scale than at congressional scale. The difference between MM and non-MM districts' PP ratios is:

- **Congressional scale**: $0.360 - 0.323 = 0.037$ PP units (10.3% penalty)
- **State house scale**: $0.387 - 0.365 = 0.022$ PP units (5.7% penalty)

The smaller compactness penalty at state house scale reflects the more granular boundary matching possible with smaller districts. At congressional scale, the algorithm must sometimes draw a non-compact district to aggregate a sufficient minority population from dispersed communities. At state house scale, individual minority communities are often large enough relative to the district target to support a majority-minority district without aggregating dispersed populations, producing a more compact result.

In all 13 states, the majority-minority districts in the state house maps achieve higher PP ratios than the majority-minority congressional districts in the same states. This confirms that the scale advantage in minority district creation is not achieved at the cost of compactness—state house majority-minority districts are both more numerous and individually more compact than their congressional counterparts.

5.4 Implications for Section 2 Compliance

The practical implication of these findings is that Section 2 compliance is *structurally easier* to achieve at state legislative scale than at congressional scale. The first *Gingles* precondition—geographic compactness sufficient to support a majority-minority district—is more readily satisfied in state house redistricting because smaller districts can match the geographic extent of minority communities more exactly. States that cannot achieve compact majority-minority congressional districts (South Carolina at congressional scale) can nevertheless achieve compact majority-minority state house districts.

This scale advantage means that states with relatively small, geographically dispersed minority populations may face Section 2 obligations at state legislative scale that they do not face at congressional scale. A minority population that is too dispersed to constitute a majority in a single congressional district may be concentrated enough, at the smaller state house scale, to constitute a majority in one or more state house districts, triggering the first *Gingles* precondition. Algorithmic redistricting at state legislative scale should therefore incorporate VRASection as a standard component, not an optional add-on.

6 Conclusion

This paper has addressed three questions about VRA compliance in algorithmic state legislative redistricting and answered each affirmatively.

First, the 42% minority population threshold identified as a practical bright line for majority-minority district formation at congressional scale holds at state house scale. VRASection applied to state house chambers in the five covered states (Alabama, Georgia, Louisiana, Mississippi, South Carolina) produces majority-minority districts using the same threshold calibration as congressional-level analysis. The threshold holds because it is a geometric property of minority population aggregation that is scale-invariant: the break between achievable and unachievable majority-minority configurations occurs at approximately the same minority population share regardless of district size. South Carolina—the one covered state that fails the VRASection test at congressional scale—succeeds at state house scale because smaller districts can match the geographic extent of its dispersed Black population concentrations.

Second, the Callais disentanglement methodology applies at state legislative scale with identical mathematical structure and qualitatively identical results. The regression of district boundary decisions on minority population share versus partisan composition produces the same pattern at state house scale as at congressional scale: large, significant coefficients on minority population share and near-zero coefficients on partisan composition, confirming that the algorithm's boundary decisions in majority-minority districts are explained by minority community geography rather than partisan strategy. This categorical satisfaction of the Callais disentanglement requirement is a consequence of the algorithm's architectural inability to access partisan data.

Third, state house maps can achieve minority opportunity district targets more efficiently than congressional maps. The comparison across 13 states shows that state house VRASection produces 7.6 times as many majority-minority districts as congressional VRASection (378 versus 50) at substantially higher compactness levels (mean PP 0.365 versus 0.323 for majority-minority districts). The compactness penalty for majority-minority districts is smaller at state house scale (5.7%) than at congressional scale (10.3%), reflecting the more granular minority community matching possible with smaller district sizes.

The broader conclusion for algorithmic redistricting practice is that VRASection should be incorporated as a standard component of state legislative redistricting pipelines, not only congressional redistricting. States with substantial minority populations that may not trigger Section 2 obligations at congressional scale may nevertheless face those obligations at state house scale, because smaller districts lower the geographic compactness barrier to majority-minority district formation. The algorithm handles this expanded scope seamlessly: VRASection's threshold-based architecture scales from 4-district congressional maps to 400-district state house maps without modification, detecting minority community concentrations at the appropriate geographic scale for the chamber being drawn.

Future work should address three open questions. First, the empirical analysis here is based on 2020 census data; the results should be validated across the 2000 and 2010 census years, particularly for states where minority population geography has changed substantially over time. Second, the analysis treats the three *Gingles* preconditions asymmetrically: we analyze precondition 1 (geographic compactness) empirically but take preconditions 2 (political cohesion) and 3 (majority bloc voting) as given. A full Section 2 analysis requires empirical analysis of actual voting data to establish cohesion and bloc voting, which is outside the scope of this paper. Third, the analysis focuses on Black and Hispanic majority-minority districts; Native American and Asian-Pacific Islander majority opportunity districts at state legislative scale are underexplored and warrant separate analysis.

The principal policy implication is encouraging: algorithmic redistricting at state legislative scale can satisfy Section 2 obligations as effectively as at congressional scale, with the additional advantage that the granular district scale allows more precise minority community boundary matching and a smaller compactness-VRA tradeoff. States adopting algorithmic redistricting for their congressional maps should consider simultaneous adoption for their state legislative maps, where the algorithm's performance advantages are, if anything, stronger.

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